



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System

COMPONENT GUIDE

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Every Reusable Component Explained

How to Read This Guide

For each component you will find:

- **What it is** — purpose in plain English
- **Props** — what inputs it accepts
- **Used in** — which pages use it
- **Why it was created** — the problem it solves

UI Components (`components/ui/`)

Button

What it is: The primary interactive element. Used for every clickable action.

Props:

Prop	Type	Default	Description
variant	string	'primary'	primary, secondary, danger, outline, ghost
size	string	'md'	sm, md, lg
isLoading	boolean	false	Shows spinner, disables button
leftIcon	node	—	Icon before label
rightIcon	node	—	Icon after label
fullWidth	boolean	false	Stretches to container width
disabled	boolean	false	Disables interaction

Used in: Every page — login forms, search actions, inventory, admin actions.

Why it was created: Without a shared Button component, every developer writes their own button HTML with different classes. This creates visual inconsistency. One Button component means one consistent style everywhere.

Badge

What it is: A small inline label used to communicate status or category.

Props:

Prop	Type	Default	Description
variant	string	'primary'	primary, secondary, success, danger, warning, info
size	string	'md'	sm, md
dot	boolean	false	Shows a coloured dot before the label

Prop	Type	Default	Description
icon	node	—	Icon before text

Example uses:

- "In Stock" — `variant="success"` dot
- "Out of Stock" — `variant="danger"` dot
- "Jan Aushadhi" — `variant="info"`
- "Admin" — `variant="danger"`

Used in: SearchResultCard, MedicineCard, UserDashboard, AdminUsers, InventoryPage.

Avatar

What it is: Displays a user's profile picture. Falls back to initials if no image.

Props:

Prop	Type	Description
src	string	Image URL
name	string	User name (used to generate initials)
size	string	xs, sm, md, lg, xl
status	string	online, offline, busy — shows status dot

Used in: TopBar (logged-in user avatar), ProfilePage, UserDashboard, AdminUsers.

Divider

What it is: A horizontal or vertical line that separates content sections.

Props:

Prop	Type	Description
orientation	string	horizontal (default) or vertical
label	string	Optional text shown in the middle of the line

Used in: Auth pages ("or continue with"), card sections, form sections.

IconButton

What it is: A square button containing only an icon. Used for compact toolbar actions.

Props:

Prop	Type	Description
icon	node	The icon to render
variant	string	Same as Button variants
size	string	sm, md, lg

Prop	Type	Description
aria-label	string	**Required** for accessibility

Used in: TopBar (theme toggle), modal close buttons, table action columns.

Form Components (`components/forms/``)

Input

What it is: A styled text input field with label, helper text, and error state.

Props:

Prop	Type	Description
label	string	Label above the input
error	string	Error message shown below
leftIcon	node	Icon inside left of input
rightIcon	node	Icon inside right of input
helperText	string	Hint text below input

Used in: LoginPage, RegisterPage, MedicineFormPage, ProfilePage.

PasswordInput

What it is: Input with a show/hide toggle for password fields.

Why it was created: Password fields need a consistent toggle behaviour. Without this component, every password field would need to duplicate the show/hide logic.

Used in: LoginPage, RegisterPage, ResetPasswordPage.

PasswordStrength

What it is: A visual bar that shows password strength (Weak / Fair / Strong / Very Strong).

How it works: Checks for: minimum length, uppercase, lowercase, number. Each criterion adds to the strength score.

Used in: RegisterPage alongside the PasswordInput.

OtpInput

What it is: Six individual single-digit input boxes for OTP entry. Automatically advances focus to the next box when a digit is entered.

Props:

Prop	Type	Description
length	number	Number of boxes (default 6)
onChange	function	Called with the full OTP string when complete

Used in: VerifyOtpPage.

Select

What it is: A styled dropdown select element.

Props:

Prop	Type	Description
options	array	[[value, label]]
label	string	Label above the select
error	string	Error message

Used in: MedicineFormPage (medicine type, category), AdminSettings.

Toggle

What it is: An on/off switch. Visually superior to a checkbox for binary settings.

Props:

Prop	Type	Description
checked	boolean	Current state
onChange	function	Called when toggled
label	string	Label beside the toggle

Used in: UserDashboard (notification settings), AdminSettings, ProfilePage.

FormField

What it is: A wrapper component that adds label, error message, and helper text to any form control.

Why it was created: Every input needs a label above and an error below. Instead of duplicating this HTML in every input component, FormField wraps any child and adds it.

Card Components ([components/cards/](#))

MedicineCard

What it is: A basic medicine display card for grids and recommendation sections.

Props:

Prop	Type	Description
medicine	object	Medicine data (name, genericName, price, availability)
onView	function	Called when card is clicked
variant	string	default, compact, horizontal

Used in: GenericRecommendationPage, UserDashboard recent medicines section.

SearchResultCard

What it is: The rich medicine card on the Search Results page. Much more detailed than MedicineCard.

Props:

Prop	Type	Description
medicine	object	Full medicine object
isComparing	boolean	Whether comparison mode is active
isSelected	boolean	Whether this card is selected for compare
onCompare	function	Toggle comparison selection
onView	function	Navigate to medicine detail
onShare	function	Share medicine (uses Web Share API)
isBestValue	boolean	Shows "Best Value" ribbon
layout	string	'grid' or 'list'

Features:

- Smart availability badges
- Compare checkbox
- Save/bookmark toggle (local state)
- Share button (Web Share API fallback)
- "Best Value" ribbon for the best-priced generic

Used in: SearchResultsPage only.

PharmacyCard

What it is: Displays a pharmacy's name, address, distance, and stock status.

Props:

Prop	Type	Description
pharmacy	object	Pharmacy data (name, address, distance, isOpen)
onView	function	Navigate to pharmacy detail
onGetDirections	function	Opens map directions

Used in: NearbyPharmaciesPage pharmacy list alongside the Leaflet map.

NotificationCard

What it is: A single notification item showing icon, message, time, and read/unread state.

Props:

Prop	Type	Description
notification	object	{ type, title, message, time, isRead }
onMarkRead	function	Mark this notification as read
onDismiss	function	Remove this notification

Used in: NotificationsPage.

InfoCard

What it is: A generic stat/info card for dashboards showing a title, value, icon, and trend.

Props:

Prop	Type	Description
title	string	Card label (e.g. "Total Searches")
value	string/number	Main metric value
icon	node	Icon to display
trend	string	up or down
trendValue	string	e.g. "+12%"
variant	string	Colour variant

Used in: UserDashboard, PharmacyDashboard, AdminDashboard.

Common Components (components/common/)

Breadcrumb

What it is: A horizontal navigation trail showing the user's location in the app hierarchy.

Example: [Home](#) > [Medicines](#) > [Paracetamol 500mg](#)

Props:

Prop	Type	Default	Description
items	array	[]	[[label, to]] — last item is current page
showHome	boolean	true	Prepend a Home link

Accessibility: Uses `<nav aria-label="Breadcrumb">`, `<ol>`, `aria-current="page"` on last item.

Used in: MedicineDetailsPage, InventoryPage, Admin pages.

SearchBar

What it is: A search input field with a submit button and optional clear button.

Props:

Prop	Type	Description
value	string	Current search value
onChange	function	Called on input change
onSubmit	function	Called on form submit
placeholder	string	Placeholder text
isLoading	boolean	Shows spinner in button

Used in: MedicineSearchPage, InventoryPage (search filter).

Pagination

What it is: Page navigation component with previous/next buttons and page number display.

Props:

Prop	Type	Description
currentPage	number	Active page
totalPages	number	Total number of pages
onPageChange	function	Called with new page number

Used in: SearchResultsPage, InventoryPage, AdminUsers, AdminMedicines.

AppErrorBoundary

What it is: A React class component that catches JavaScript errors in the component tree and shows [ServerErrorPage](#) instead of a blank screen.

Why it was created: Without an error boundary, any uncaught JavaScript error crashes the entire app and shows a blank white screen. This catches it gracefully.

Used in: `main.jsx` — wraps the entire application.

Feedback Components (`components/feedback/`)

Spinner

Props: `size` (sm/md/lg), `color`, `label` (screen reader text)

Used in: Route loading fallback, button loading states, page loading states.

Skeleton

What it is: A grey animated placeholder that mimics the shape of content while it loads. Better UX than a spinner because users can see the layout before data arrives.

Used in: Medicine search results while loading, dashboard sections.

EmptyState

What it is: A friendly "nothing to show" screen with an icon and description.

Props: `title`, `description`, `icon`, `actionLabel`, `onAction`

Used in: Search results (no medicines found), notifications (no notifications), inventory (empty stock).

ErrorState

What it is: A friendly "something went wrong" screen with a retry button.

Props: `title`, `description`, `onRetry`

Used in: Any data-loading section that might fail.

Navigation Components (`components/navigation/`)

Navbar

What it is: The top navigation bar on the public home page. Shows the app logo, nav links, and login/register buttons.

Used in: `MainLayout` (home page only).

Sidebar

What it is: The left navigation panel in all authenticated dashboards. Renders nav items from `navConfig.js` based on the current role.

How it works: Reads the nav items array (`USER_NAV`, `PHARMACY_NAV`, `ADMIN_NAV`) and renders links. Active link is highlighted using React Router's `NavLink`.

Used in: `UserLayout`, `PharmacyLayout`, `AdminLayout`.

TopBar

What it is: The horizontal top bar inside authenticated layouts. Shows page title, user avatar, notification bell, and theme toggle.

Used in: `UserLayout`, `PharmacyLayout`, `AdminLayout`.

Footer

What it is: The site footer with project information, links, and copyright.

Used in: [MainLayout](#) (public home page).

Dialog Components (`components/dialogs/`)

Modal

What it is: A configurable overlay dialog.

Props:

Prop	Type	Description
isOpen	boolean	Whether the modal is visible
onClose	function	Called when closing
title	string	Modal header title
size	string	sm, md, lg, xl

Accessibility: Traps focus inside the modal when open, closes on Escape key.

Used in: [MedicineDetailsPage](#) (quick view), [GenericRecommendationPage](#).

ConfirmDialog

What it is: A specialised modal with "Confirm" and "Cancel" buttons. Used for destructive actions.

Props:

Prop	Type	Description
isOpen	boolean	Visibility
onConfirm	function	Called on confirm
onCancel	function	Called on cancel
title	string	Dialog heading
description	string	Explanation of what will happen
confirmLabel	string	Button text (default "Confirm")
isDangerous	boolean	Makes confirm button red

Used in: [InventoryPage](#) (delete stock), [AdminUsers](#) (delete user), [AdminMedicines](#) (delete medicine).

Layout Components (`components/layout/`)

Container

What it is: A `<div>` with `max-width: 1280px` and responsive horizontal padding. Prevents content from stretching too wide on large screens.

Used in: Every page section.

PageHeader

What it is: A consistent page heading block with title, optional subtitle, breadcrumb, and action button.

Props:

Prop	Type	Description
title	string	Page title
subtitle	string	Page subtitle
breadcrumbs	array	Breadcrumb items
action	node	Optional action button

Used in: InventoryPage, AdminPages, ProfilePage.

SectionHeader

What it is: A section title within a page (h2 level).

Used in: HomePage sections, UserDashboard sections.